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Geometry of classical groups over finite fields and its applications

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Abstract

This is a survey paper on the geometry of classical groups over finite fields and its applications, old and new. The contents of the paper are as follows:

(1) The seed of our study; (2) The problems we are interested in; (3) History; (4) Recent results; (5) Application to association schemes and designs; (6) Application to authentication codes; (7) Application to projective codes; (8) Application to lattices generated by orbits of subspaces; (9) Application to representations of forms by forms; (10) Concluding remarks.

1. The seed of our study

Let $\mathbb{F}_q$ be a finite field with q elements, where q is a prime power, $\mathbb{F}_q^{(n)}$ be the n-dimensional row vector space over $\mathbb{F}_q$, and $\mathrm{GL}_n(\mathbb{F}_q)$ be the *general linear group* of degree n over $\mathbb{F}_q$. $\mathrm{GL}_n(\mathbb{F}_q)$ acts on $\mathbb{F}_q^{(n)}$ in the following way:

$$\begin{aligned} &\mathbb{F}_q^{(n)} \times \mathrm{GL}_n(\mathbb{F}_q) \to \mathbb{F}_q^{(n)}, \\ &((x_1, x_2, \ldots, x_n), T) \mapsto (x_1, x_2, \ldots, x_n)T. \end{aligned} \tag{1}$$

Let P be an m-dimensional subspace of $\mathbb{F}_q^{(n)}$ and $v_1, v_2, \ldots, v_m$ be a basis of P, then

$$\begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_m \end{pmatrix} \tag{2}$$

is an $m \times n$ matrix of rank m over $\mathbb{F}_q$. We call the matrix (2) a *matrix representation* of the subspace P and use also the same letter P to denote the matrix (2) if no ambiguity arises. The action (1) of $\mathrm{GL}_n(\mathbb{F}_q)$ on $\mathbb{F}_q^{(n)}$ induces an action on the set of subspaces

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of $\mathbb{F}_q^{(n)}$ such that $T \in \mathrm{GL}_n(\mathbb{F}_q)$ carries the subspace P to PT. We may propose the following problems:

(i) What are the orbits of subspaces of $\mathbb{F}_q^{(n)}$ under the action of $\mathrm{GL}_n(\mathbb{F}_q)$?

(ii) How many orbits are there?

(iii) What are the lengths of the orbits?

(iv) What is the number of subspaces in an orbit contained in a given subspace?

The answers to these four problems are well known, they are:

(i) Two subspaces belong to the same orbit if and only if their dimensions are equal.

(ii) There are altogether $n+1$ orbits.

(iii) Denote the length of the orbit of m-dimensional subspaces $(0 \leqslant m \leqslant n)$ by $N(m,n)$, then

$$N(m,n) = \frac{\prod_{i=n-m+1}^{n} (q^i - 1)}{\prod_{i=1}^{m} (q^i - 1)}. \tag{3}$$

(iv) The number of k-dimensional subspaces contained in a given m-dimensional subspace $(0 \leqslant k \leqslant m \leqslant n)$ is $N(k,m)$.

2. The problems we are interested in

It is natural to propose the following problems:

Use any one of the other classical groups, such as the symplectic group $\mathrm{Sp}_n(\mathbb{F}_q)$ (where $n = 2v$), the unitary group $U_n(\mathbb{F}_q)$ (where q is a square) and the orthogonal group $O_n(\mathbb{F}_q)$ (where $n = 2v + \delta$ and $\delta = 0, 1,$ or 2) to replace $\mathrm{GL}_n(\mathbb{F}_q)$, then study problems (i)–(iv).

The above problems look quite classical. For instance, they could have been proposed in 1900, the year Dickson published his famous monograph *Linear Groups* [16], in which all the ingredients for proposing them are contained. But as far as I know, mathematicians started to study these problems much later.

3. The history of the problems

In 1937 Witt [72] studied the above mentioned problem (i) for the orthogonal group over any field F of characteristic $\neq 2$. Let S be an $n \times n$ nonsingular symmetric matrix over F. The *orthogonal group* of degree n over F relative to S, denoted by $O_n(F,S)$, is defined to be

$$O_n(F,S) = \{T \in \mathrm{GL}_n(F) \mid TS\,{}^{\mathrm{t}}T = S\}. \tag{4}$$

The famous Witt's theorem asserts that two subspaces P_1 and P_2 of $F^{(n)}$ belong to the same orbit of $O_n(F,S)$ if and only if $\dim P_1 = \dim P_2$ and $P_1 S\,{}^{\mathrm{t}}P_1$ and $P_2 S\,{}^{\mathrm{t}}P_2$ are cogredient. Later Witt's theorem was generalized to other classical groups by Arf [1],

Dieudonné [17,18], Hua [26], etc. [14,15,38]. It is worth to mention that Hua [26] gave also a short and elegant matrix proof of the generalized Witt's theorem, which was reproduced in [59].

In his invited address at the International Congress of Mathematicians, Edinburgh, 1958, Segre [43] announced his formula for the number of $(m-1)$-dimensional flats lying on a nondegenerate quadric in $\mathrm{PG}(n-1,\mathbb{F}_q)$, the $(n-1)$-dimensional projective space over $\mathbb{F}_q$:

$$\frac{\prod_{i=v-m+1}^{v}(q^i-1)(q^{i+\delta-1}-1)}{\prod_{i=1}^{m}(q^i-1)}, \tag{5}$$

where $1\leqslant m\leqslant v$, $v=(n-1)/2$ and $\delta=1$ when n is odd, and $v=n/2$, $\delta=0$ or $v=n/2-1$, $\delta=2$ when n is even and the quadric is of the hyperbolic type or the elliptic type, respectively. The complete proof of this formula appeared one year later [44]. He used a geometric method to deduce this formula, which works for both the case of characteristic not two and the case of characteristic two. In 1962 Ray-Chaudhuri [41] also deduced Segre's formula (5).

For simplicity, assume that q is odd. Then the $(m-1)$-dimensional flats lying on a nondegenerate quadric in $\mathrm{PG}(n-1,\mathbb{F}_q)$ are m-dimensional totally isotropic subspaces of $\mathbb{F}_q^{(n)}$ with respect to the $n\times n$ nonsingular symmetric matrix S defining the quadric, which, by Witt's theorem, form an orbit of subspaces under the orthogonal group $O_n(\mathbb{F}_q,S)$. Thus, Segre actually studied the above-mentioned problem (iii) for the orthogonal group over $\mathbb{F}_q$, but he restricted himself to consider only the orbits of totally isotropic or totally singular subspaces corresponding to cases when q is odd or even, respectively.

In 1964 Dai, Feng, Yang, three students of mine at that time and myself [15,20,47,69] studied problem (iii) for the groups $\mathrm{Sp}_{2v}(\mathbb{F}_q)$, $U_n(\mathbb{F}_q)$ (where q is a square), and $O_{2v+\delta}(\mathbb{F}_q)$ (where $\delta=0,1$, or 2). We determined not only the lengths of those orbits of totally isotropic or totally singular subspaces but also the lengths of all the orbits. We call these results the Anzahl theorems in the geometries of these classical groups. Our methods were algebraic and our results were compiled in our monograph [67]. In 1965 Pless [39] computed the lengths of the orbits of totally isotropic subspaces of $\mathbb{F}_q^{(2v)}$ under the group $\mathrm{Sp}_{2v}(\mathbb{F}_q)$ and the number of totally isotropic subspaces of the same dimension of $\mathbb{F}_q^{(2v+\delta)}$ (where $\delta=1$ or 2) with respect to a $(2v+\delta)\times(2v+\delta)$ nonsingular non-alternate symmetric matrix over $\mathbb{F}_q$ when q is even. In 1965 Segre [45] and in 1966 Bose and Chakravarti [3] determined the lengths of the orbits of totally isotropic subspaces of $\mathbb{F}_q^{(n)}$ under the group $U_n(\mathbb{F}_q)$ (where q is a square).

In 1966 the author studied problem (iv) for the group $\mathrm{Sp}_{2v}(\mathbb{F}_q)$, $U_n(\mathbb{F}_q)$ (q is a square), and $O_{2v+\delta}(\mathbb{F}_q)$ (where $\delta=0,1$, or 2) and obtained closed formulas for the number of subspaces in an orbit under each of these groups contained in a given subspace. These results are also called Anzahl theorems and were compiled in [67] too.

4. Recent results

In the early 1990s I returned to the study of the geometry of classical groups over finite fields and obtained the following results:

(1) Problems (i) and (ii) for the symplectic, unitary, and orthogonal groups over finite fields are studied [51,54,60,61]. Of course, Witt's theorem and its generalizations give a solution to problem (i), but we would like to use a set of numerical invariants to characterize an orbit and to derive the conditions satisfied by them when such an orbit exists, then the number of orbits can be computed.

Take the symplectic case as an example. Let

$$K = \begin{pmatrix} 0 & I^{(v)} \\ -I^{(v)} & 0 \end{pmatrix}. \tag{6}$$

Then the symplectic group of degree $2v$ is defined as

$$\mathrm{Sp}_{2v}(\mathbb{F}_q) = \{T \in \mathrm{GL}_{2v}(\mathbb{F}_q) \mid TK\,{}^tT = K\}. \tag{7}$$

Let P be an m-dimensional subspace of $\mathbb{F}_q^{(2v)}$. Clearly, $PK\,{}^tP$ is alternate and, hence, the rank of $PK\,{}^tP$ is even. Assume that rank of $PK\,{}^tP = 2s$, then P is said to be of *type* (m,s). From Dieudonné's generalization [17] of Witt's theorem it follows that two subspaces belong to the same orbit under $\mathrm{Sp}_{2v}(\mathbb{F}_q)$ if and only if they are of the same type. It can be proved [51] that the type (m,s) of a subspace satisfies the inequality

$$2s \leqslant m \leqslant v + s \tag{8}$$

and that for any pair of nonnegative integers (m,s) satisfying (8) there exist subspaces of type (m,s). Thus, the number of orbits of subspaces under $\mathrm{Sp}_{2v}(\mathbb{F}_q)$ is equal to the number of pairs of nonnegative integers (m,s) satisfying (8). We computed that the latter is equal to

$$\tfrac{1}{2}(v+1)(v+2). \tag{9}$$

By the way we mention that the length $N(m,s;2v)$ of the orbit of subspaces of type (m,s) of $\mathbb{F}_q^{(2v)}$ given in [47] is

$$N(m,s;2v) = q^{2s(v+s-m)} \frac{\prod_{i=v+s-m+1}^{v} (q^{2i}-1)}{\prod_{i=1}^{s} (q^{2i}-1)} \prod_{i=1}^{m-2s} (q^i - 1). \tag{10}$$

This is the solution to problem (iii) for the symplectic group.

(2) The singular symplectic, unitary, and orthogonal groups are introduced and the problems (i)–(iv) are studied [62,63].

Take the singular symplectic case as an example. Let

$$K_l = \begin{pmatrix} K & \\ & 0^{(l)} \end{pmatrix}, \tag{11}$$

where K is the nonsingular alternate matrix (6). Define

$$\mathrm{Sp}_{2v+l,v}(\mathbb{F}_q) = \{T \in \mathrm{GL}_{2v+l}(\mathbb{F}_q) \mid TK_l\,{}^tT = K_l\}, \tag{12}$$

which is called the *singular symplectic group* over $\mathbb{F}_q$. Clearly, $\mathrm{Sp}_{2v+l,v}(\mathbb{F}_q)$ acts on $\mathbb{F}_q^{(2v+l)}$ in an obvious way. Then problems (i)–(iv) can be studied for $\mathrm{Sp}_{2v+l,v}(\mathbb{F}_q)$, and complete results are obtained.

Similarly, singular unitary and orthogonal groups over $\mathbb{F}_q$ can be defined, and complete results for problems (i)–(iv) are obtained.

A natural question arises. Why do we study the geometry of singular symplectic, unitary, and orthogonal groups over finite fields?

The answer to problem (iv) for the general linear group $\mathrm{GL}_n(\mathbb{F}_q)$ is easy: the number of k-dimensional subspaces contained in a given m-dimensional subspace $(0 \leqslant k \leqslant m \leqslant n)$ of $\mathbb{F}_q^{(n)}$ is $N(k,m)$. However, problem (iv) for the other classical groups is not so easy.

Take again the symplectic case as an example. Now assume that $\mathrm{Sp}_{2v}(\mathbb{F}_q)$ acts on $\mathbb{F}_q^{(2v)}$. Given a subspace P of type (m,s), where (m,s) satisfies (8), we would like to compute the number of subspaces of type (m_1,s_1), where $2s_1 \leqslant m_1 \leqslant v + s_1$, contained in P. Denote this number by $N(m_1,s_1;m,s;2v)$. We may choose a matrix representation of P, denoted by P again, such that

$$PK\,{}^tP = \begin{pmatrix} 0 & I^{(s)} & \\ -I^{(s)} & 0 & \\ & & 0^{(m-2s)} \end{pmatrix}. \tag{13}$$

Let P_1 be a subspace of type (m_1,s_1) contained in P. As an m_1-dimensional subspace of the m-dimensional space P, P_1 has a matrix representation, denoted by P_1 again, which is an $m_1 \times m$ matrix of rank m_1. Then as a subspace of $\mathbb{F}_q^{(2v)}$, the subspace P_1 has P_1P as a matrix representation. Similarly, we can choose the matrix P_1 such that

$$(P_1P)K\,{}^t(P_1P) = \begin{pmatrix} 0 & I^{(s_1)} & \\ -I^{(s_1)} & 0 & \\ & & 0^{(m_1-2s_1)} \end{pmatrix}. \tag{14}$$

Then

$$P_1 \begin{pmatrix} 0 & I^{(s)} & \\ -I^{(s)} & 0 & \\ & & 0^{(m-2s)} \end{pmatrix} {}^tP_1 = \begin{pmatrix} 0 & I^{(s_1)} & \\ -I^{(s_1)} & 0 & \\ & & 0^{(m_1-2s_1)} \end{pmatrix}. \tag{15}$$

Thus, for any $T \in \mathrm{Sp}_{2s+(m-2s),s}(\mathbb{F}_q)$, P_1TP is also a matrix representation of a subspace of type (m_1,s_1) and contained in P and as a subspace of P it is represented by the matrix P_1T. Therefore, it is natural to introduce the singular symplectic group $\mathrm{Sp}_{2s+(m-2s),s}(\mathbb{F}_q)$ and study how the subspaces of $\mathbb{F}_q^{(m)}$ are subdivided into orbits under $\mathrm{Sp}_{2s+(m-2s),s}(\mathbb{F}_q)$, the length of each orbit, and what orbits are contained in P.

(3) For pseudo-symplectic groups over finite fields of characteristic 2 problems (i)–(iv) are also studied [36,56].

Now let $\mathbb{F}_q$ be a finite field of characteristic 2, then any $n \times n$ nonsingular non-alternate symmetric matrix over $\mathbb{F}_q$ is cogredient to either

$$S_1 = \begin{pmatrix} 0 & I^{(v)} & \\ I^{(v)} & 0 & \\ & & 1 \end{pmatrix} \quad \text{when } n = 2v+1 \text{ is odd} \tag{16}$$

or

$$S_2 = \begin{pmatrix} 0 & I^{(v)} & & \\ I^{(v)} & 0 & & \\ & & 0 & 1 \\ & & 1 & 1 \end{pmatrix} \quad \text{when } n = 2v+2 \text{ is even.} \tag{17}$$

We use S_δ ($\delta = 1$ or 2) to cover these two cases. Define the *pseudo-symplectic group* of degree $2v+\delta$ over $\mathbb{F}_q$ to be

$$\mathrm{Ps}_{2v+\delta}(\mathbb{F}_q) = \{T \in \mathrm{GL}_{2v+\delta}(\mathbb{F}_q) \mid TS_\delta {}^tT = S_\delta\}. \tag{18}$$

It was proved by Dieudonné [17] that $\mathrm{Ps}_{2v+1}(\mathbb{F}_q) \simeq \mathrm{Sp}_{2v}(\mathbb{F}_q)$ and $\mathrm{Ps}_{2v+2}(\mathbb{F}_q)$ has a normal series with $\mathrm{Sp}_{2v}(\mathbb{F}_q)$ as one of its factors and with the additive group of $\mathbb{F}_q$ as all the other factors. Thus, from a group theory point of view the pseudo-symplectic group $\mathrm{Ps}_{2v+\delta}(\mathbb{F}_q)$ is less interesting. However, its geometry is very peculiar. Let P be an m-dimensional subspace of $\mathbb{F}_q^{(2v+\delta)}$, then $PS_\delta {}^tP$ is a symmetric matrix and is cogredient to one of the following normal forms:

$$\begin{pmatrix} 0 & I^{(s)} & \\ I^{(s)} & 0 & \\ & & 0^{(m-2s)} \end{pmatrix}, \tag{19}$$

$$\begin{pmatrix} 0 & I^{(s)} & & \\ I^{(s)} & 0 & & \\ & & 1 & \\ & & & 0^{(m-2s-1)} \end{pmatrix}, \tag{20}$$

and

$$\begin{pmatrix} 0 & I^{(s)} & & & \\ I^{(s)} & 0 & & & \\ & & 0 & 1 & \\ & & 1 & 1 & \\ & & & & 0^{(m-2s-2)} \end{pmatrix}. \tag{21}$$

P is called a subspace of *type* $(m, 2s+\tau, s, \varepsilon)$, where $\tau = 0, 1$, or 2 corresponding to the above three normal forms (19), (20), or (21), respectively, $\varepsilon = 0$ or 1 corresponding to the cases $e_{2v+1} \notin P$ or $e_{2v+1} \in P$, respectively, and e_{2v+1} is the $(2v+\delta)$-dimensional row vector whose $(2v+1)$th component is 1 and other components are all 0's. It is proved [36] that two subspaces of $\mathbb{F}_q^{(2v+\delta)}$ belong to the same orbit under $\mathrm{Ps}_{2v+\delta}(\mathbb{F}_q)$ if

and only if they are of the same type and that subspaces of type $(m, 2s+\tau, s, \varepsilon)$ exist if and only if

$$(\tau,\varepsilon)=\begin{cases}(0,0),(1,0),(1,1), \text{ or } (2,0), & \text{when } \delta=1,\\ (0,0),(0,1),(1,0),(2,0), \text{ or } (2,1), & \text{when } \delta=2\end{cases} \tag{22}$$

and

$$2s+\max\{\tau,\varepsilon\}\leqslant m\leqslant v+s+[(\tau+\delta-1)/2]+\varepsilon. \tag{23}$$

Using conditions (22) and (23) we can compute the number of orbits of subspaces under $\mathrm{Ps}_{2v+\delta}(\mathbb{F}_q)$, which is equal to

$$\tfrac{1}{2}(v+1)((v+4)\delta+3v). \tag{24}$$

Denote the length of the orbit of subspaces of type $(m, 2s+\tau, s, \varepsilon)$ by $N(m, 2s+\tau, s, \varepsilon; 2v+\delta)$. Then

$$N(m,2s+\tau,s,\varepsilon;2v+\delta)=q^{n_0+2(s+(2-\delta)[\tau/2])(v+s-m+\delta[(\tau+1)/2]+(\delta-1)(\tau-1)(\tau-2)\varepsilon/2)}\times\frac{\prod_{i=v+s-m+[(\tau+\delta-1)/2]+\varepsilon+1}^{v}(q^{2i}-1)}{\prod_{i=1}^{s}(q^{2i}-1)\prod_{i=1}^{m-2s-\max(\tau,\varepsilon)}(q^i-1)}, \tag{25}$$

where $n_0=1$ when $\delta=1$, and $n_0=m, 0, 2(v+1)-m$, $2(v+1)-m$, or $2(v+1)-m$ corresponding to the cases $(\tau,\varepsilon)=(0,0),(0,1),(1,0),(2,0)$, or $(2,1)$, respectively, when $\delta=2$.

In order to study problem (iv) for the pseudo-symplectic group $\mathrm{Ps}_{2v+\delta}(\mathbb{F}_q)$ the singular pseudo-symplectic group is introduced and for which problems (i)–(iii) are studied [56].

(4) The affine classification of quadrics over finite fields is obtained [52,53].

The foregoing results together with our results obtained in the mid-1960s are compiled in a monograph [64].

5. Application to association schemes and designs

In 1952 Bose and Shimamato [4] separated the concept of association schemes from the concept of PBIB designs. Around 1960 Hsu emphasized the importance of association schemes in various occasions in China. He also made important contributions to association schemes [2]; for instance, in 1964 he defined the association scheme of the Grassmann manifold of the m-dimensional subspaces in $\mathbb{F}_q^{(n)}$ $(1\leqslant m<n)$ by defining two m-dimensional subspaces P and Q to be the ith associates if $\dim(P+Q)=m+i$ $(i=1,2,\ldots,\min\{m,n-m\})$ and computed its parameters v, n_i, p_{jk}^i $(i,j,k=1,2,\ldots,\min\{m,n-m\})$. This is one of the earliest nontrivial association schemes besides the known strongly regular graphs in the early sixties. Following Hsu, in 1965 the author [49] used a group theoretical method to compute the parameters of this association scheme which seems to be simpler, and in 1965–1966 Dai, Feng, Yang, and the

author [67,74,75] defined association schemes of the maximal totally isotropic (or singular) subspaces in the symplectic, unitary, and orthogonal geometries over $\mathbb{F}_q$ (which was later called the dual polar spaces of type C_v, ${}^2A_{2v}$, ${}^2A_{2v-1}$, B_v, D_v, and ${}^2D_{v+1}$ [5]) in the same way as the association scheme of the Grassmann manifold defined by Hsu and computed their parameters v, n_i, p_{jk}^i. The parameters v are given by the Anzahl theorems, and the parameters n_i and p_{jk}^i are computed by the group-theoretical method. All these results were complied in the monograph [67].

In 1954 Clatworthy [13] showed that a geometric configuration in $\mathrm{PG}(3,\mathbb{F}_q)$ may be interpreted as a PBIB design. In our terminology, he took the set of one-dimensional subspaces of the four-dimensional symplectic space over $\mathbb{F}_q$ as the set of points and the set of two-dimensional totally isotropic subspaces as the set of blocks. Two points are said to be the first associates (or second associates) if they span a two-dimensional totally isotropic subspace (or nonisotropic subspace), respectively. A point is defined to lie in a block if the one-dimensional subspace of the point is contained in the two-dimensional totally isotropic subspace as the block. Then a PBIB(2) design is obtained. Clatworthy also computed the parameters of the design.

In 1962 Ray-Chaudhuri [42] used the geometry of orthogonal groups, which was called the geometry of quadrics by him, to construct PBIB designs. He constructed several PBIB(2) designs with one-dimensional or two-dimensional totally isotropic (or singular) subspaces of the geometry of orthogonal groups over finite fields as points or blocks and computed their parameters. At that time only the length of the orbit of totally isotropic (or singular) subspaces of a given dimension under the orthogonal group over finite fields was known, so he naturally restricted himself to take only totally isotropic (or singular) subspaces as points and blocks in order to compute the parameters of the designs he constructed.

In the mid-1960s after we had found the closed formulas for the lengths of all the orbits of subspaces under the symplectic, unitary, and orthogonal groups over finite fields, we [48,67,69,74,75] constructed many PBIB designs by taking the one-dimensional, two-dimensional, or v-dimensional totally isotropic (or singular) subspaces in the geometries of these classical groups as points and subspaces of any given type as blocks and computed their parameters. All these results were also compiled in the monograph [67] and were sketched in [55].

However, we would like to mention that when we took the two-dimensional totally isotropic (or singular) subspaces as points to construct association schemes and PBIB designs, we consider only the symplectic, unitary, and orthogonal spaces of low dimensions, because it gave us PBIB (2) designs. Of course, we can take any orbit of subspaces under the symplectic, unitary, or orthogonal group over finite fields as the set of points and define the associate relation according to the orbit of pairs of points under that group, then an association scheme is obtained. Moreover, if we take any orbit of subspaces as the set of blocks and define a point to lie in a block in a certain way, then a PBIB design is obtained. In a short note [50] published in 1965 some association schemes and PBIB designs were constructed by taking the one-dimensional nonisotropic (or non-singular) subspaces in the unitary or orthogonal geometry over

some small fields as points and their parameters were computed. In the 1980s the ideas of taking the one-dimensional nonisotropic (or nonsingular) subspaces and taking the two-dimensional totally isotropic (or singular) subspaces as points were carried out and generalized by several Chinese mathematicians and their works were sketched in [55] and will not be repeated here.

6. Application to authentication codes

The first authentication code, constructed by Gilbert et al. [23] in 1974, were based on the projective geometry over finite fields. Projective geometry, according to Klein's Erlangen Program, is the geometry of the projective general linear group. Then it is natural to propose the problem whether it is possible to construct authentication codes from the geometry of symplectic, unitary, or orthogonal groups over finite fields. The answer is of course positive and some authentication codes have been so constructed [19,57,58,68]. To illustrate we give a construction of Feng and Wan [19] below.

Consider the $2v$-dimensional symplectic space over $\mathbb{F}_q$, i.e., the $2v$-dimensional row vector space $\mathbb{F}_q^{(2v)}$ on which the symplectic group $\mathrm{Sp}_{2v}(\mathbb{F}_q)$ acts. Assume that $v \geqslant 2$ and let s be an integer such that $1 \leqslant s < v$. Let P_0 be a fixed subspace of type $(s,0)$. Take the set of subspaces of type $(2s,s)$ containing P_0 to be the set $\mathscr{S}$ of source states, the set of s-dimensional subspaces whose joins with P_0 are subspaces of type $(2s,s)$ to be the set $\mathscr{E}$ of encoding rules and also the set $\mathscr{M}$ of messages. For any source state s and encoding rule e, let $f(s,e) = s \cap e^{\perp}$, where

$$e^{\perp} = \{x \in \mathbb{F}_q^{(2v)} \mid xK\,{}^{\mathrm{t}}e = 0\}. \tag{26}$$

It can be proved that $s \cap e^{\perp}$ is an s-dimensional subspace whose join with P_0 is of type $(2s,s)$. Thus, we may define $f(s,e) = s \cap e^{\perp}$ to be the message into which the source state s is encoded using the encoding rule e. Then a Cartesian authentication code is obtained and its size parameters are

$$|\mathscr{S}| = q^{2s(v-s)}, \qquad |\mathscr{E}| = |\mathscr{M}| = q^{s(2v-s)}. \tag{27}$$

Now assume that the encoding rules are chosen according to a uniform probability distribution. Then the probabilities of a successful impersonation attack and a successful substitution attack are, respectively,

$$P_{\mathrm{I}} = \frac{1}{q^{s^2}}, \qquad P_{\mathrm{S}} = \frac{1}{q^s}. \tag{28}$$

In virtue of the combinatorial lower bounds $P_{\mathrm{I}} \geqslant |\mathscr{S}|/|\mathscr{M}|$ and $P_{\mathrm{S}} \geqslant (|\mathscr{S}| - 1)/(|\mathscr{M}| - 1)$, for the authentication code constructed above P_{I} is optimal. If we require the order of magnitude of P_{S} as a function of q to be optimal, then for the code, P_{S} is nearly optimal when and only when $s = 1$.

Similar constructions can be done for the unitary and orthogonal cases.

7. Application to projective codes

Let C be a linear $[n,k]$-code over $\mathbb{F}_q$. C is called a *projective code* if the columns of a generator matrix of C are not proportional.

Let Q be a $k \times k$ matrix over $\mathbb{F}_q$. The set of points ${}^t(x_1,x_2,\ldots,x_k)$ in $\mathrm{PG}(k-1,\mathbb{F}_q)$ satisfying

$$(x_1,x_2,\ldots,x_k)Q\,{}^t(x_1,x_2,\ldots,x_k) = 0 \tag{29}$$

is called a *quadric* in $\mathrm{PG}(k-1,\mathbb{F}_q)$ and will also be denoted by Q. The number of points of Q will be denoted by $|Q|$ and can be derived from the Anzahl theorems of the singular orthogonal geometry over $\mathbb{F}_q$. In particular, if Q is nondegenerate, i.e., (29) cannot be carried under projective transformations into a quadratic homogeneous equation with less than k indeterminates, then $|Q|$ was obtained by Primrose [40] and is also contained in Segre's formula (5).

Now let Q be a quadric in $\mathrm{PG}(k-1,\mathbb{F}_q)$ and assume that $|Q|=n$. For each point Q choose a system of coordinates and regard it as a k-dimensional column vector. Arrange these n column vectors in any order into a $k \times n$ matrix, denoted by G_Q. It can be proved that G_Q is of rank k. Hence, G_Q can be regarded as a generator matrix of a projective $[n,k]$-code, which is denoted by C_Q and called the projective code from the quadric Q in $\mathrm{PG}(k-1,\mathbb{F}_q)$.

In 1975 Wolfmann [73] proved that if Q is nondegenerate, then C_Q has only two or three distinct nonzero weights when k is even or odd, respectively, by computing the values of the nonzero weights of C_Q, and he computed also their multiplicities.

Let C be a linear $[n,k]$-code over $\mathbb{F}_q$. For a linear subcode D of C, let $\chi(D)$ be the *support* of D, namely,

$$\chi(D) = \{i \mid x_i \neq 0 \text{ for some } (x_1,x_2,\ldots,x_n) \in D\}. \tag{30}$$

The rth *minimum support weight* [24] or *generalized Hamming weight* [70] of C, denoted by $d_r(C)$, is then defined as

$$d_r(C) = \min\{|\chi(D)| \mid D \text{ is an } r\text{-dimensional subcode of } C\}. \tag{31}$$

Obviously, $d_1(C)$ is just the minimum Hamming weight of the code C. The *weight hierarchy* of C is then defined to be the set of generalized Hamming weights

$$\{d_1(C), d_2(C), \ldots, d_k(C)\}. \tag{32}$$

Using our results on the geometry of orthogonal groups over finite field, the author [65] computed the weight hierarchies of the q-ary projective codes from nondegenerate quadrics in projective spaces.

The corresponding problems for the projective codes from nonsingular Hermitian varieties in projective spaces over $\mathbb{F}_{q^2}$ were studied by Chakravarti [9], and Hirschfeld et al. [25]. More precisely, the former proved that the code has only two distinct nonzero weights and computed the values and multiplicities of them, and the latter three computed the weight hierarchies of the code.

A linear $[n,k]$-code C is said to satisfy the *chain condition* [71], if there exist r-dimensional subcodes D_r of C for $1 \leqslant r \leqslant k$ such that

$$|\chi(D_r)| = d_r(C), \quad r = 1,2,\ldots,k, \tag{33}$$

and

$$D_1 \subseteq D_2 \subseteq \cdots \subseteq D_k. \tag{34}$$

Using the geometry of orthogonal groups over finite fields, the author [65] proved that the projective codes C_Q from nondegenerate quadrics Q in $\mathrm{PG}(k-1, \mathbb{F}_q)$ satisfy the chain condition.

8. Application to lattices generated by orbits of subspaces

Let G_n denote any one of the classical groups $\mathrm{GL}_n(\mathbb{F}_q)$, $\mathrm{Sp}_{2v}(\mathbb{F}_q)$ $(n=2v)$, $U_n(\mathbb{F}_q)$ $(q=q_0^2)$, $O_{2v+\delta}(\mathbb{F}_q)$ $(n=2v+\delta,\ \delta=0,\ 1,\ \text{or } 2)$, and $Ps_{2v+\delta}(\mathbb{F}_q)$ (q even, $n=2v+\delta$, $\delta=1$ or 2), $\mathscr{M}$ be any orbit of subspaces of $\mathbb{F}_q^{(n)}$ under G_n, and $\mathscr{L}$ be the set of intersections of elements of $\mathscr{M}$. Partially order $\mathscr{L}$ by the reverse inclusion, then $\mathscr{L}$ is a finite geometric lattice with $\mathbb{F}_q^{(n)}$ as its minimal element, $\mathscr{M}$ as its set of atoms, and $r(X) = \dim(\mathbb{F}_q^{(n)}/X)$ for any $X \in \mathscr{L}$ as its rank function. $\mathscr{L}$ is called the lattice generated by $\mathscr{M}$ and the characteristic polynomial of $\mathscr{L}$ is defined by

$$\chi(\mathscr{L}, t) = \sum_{X \in \mathscr{L}} \mu(\mathbb{F}_q^{(n)}, X) t^{\dim X}, \tag{35}$$

where μ is the Möbius function of $\mathscr{L}$. For any fixed G_n, the following problems can be proposed. (i) What are the inclusion relations between the different $\mathscr{L}$'s? (ii) What are the conditions for a subspace belonging to a given $\mathscr{L}$? (iii) The determination of the characteristic polynomials of all the $\mathscr{L}$'s. When $G_n = O_{2v+\delta}(\mathbb{F}_q)$ or $\mathrm{Ps}_{2v+\delta}(\mathbb{F}_q)$, let $\mathscr{M}'$ be any set of subspaces of the same dimension and rank. For $O_{2v+\delta}(\mathbb{F}_q)$, q being odd, let S be the $(2v+\delta) \times (2v+\delta)$ nonsingular symmetric matrix defining the group $O_{2v+\delta}(\mathbb{F}_q)$, the rank of a subspace P of $\mathbb{F}_q^{(2v+\delta)}$ is defined to be rank PS^tP. For $O_{2v+\delta}(\mathbb{F}_q)$ and $\mathrm{Ps}_{2v+\delta}(\mathbb{F}_q)$, q being even, the rank of a subspace can be defined in a similar way. Let $\mathscr{L}'$ be the set of intersections of elements of $\mathscr{M}'$. Partially order $\mathscr{L}'$ by the reverse inclusion, then $\mathscr{L}'$ is also a finite-geometric lattice and the same problems (i)–(iii) can be proposed for the $\mathscr{L}'$s.

In 1985 Orlik and Solomon [37] gave the solutions to problems (ii) and (iii) for the case when $G_n = U_n(\mathbb{F}_q)$ $(q=q_0^2)$ and $\mathscr{M}$ is the orbit of $(n-1)$-dimensional nonisotropic subspaces and for the case when $G_n = O_{2v+\delta}(\mathbb{F}_q)$, ($q$ odd), and $\mathscr{M}'$ is the set of $(n-1)$-dimensional nonisotropic subspaces. In 1991 and 1993, respectively, Chen and Wan [10,11] gave the solutions to problems (ii) and (iii) for the case when $G_n = \mathrm{Sp}_{2v}(\mathbb{F}_q)$ and $\mathscr{M}$ is the orbit of $(2v-2)$-dimensional nonisotropic subspaces and for the case when $G_n = O_{2v+\delta}(\mathbb{F}_q)$, ($q$ even), and $\mathscr{M}'$ is the set of $(n-1)$-dimensional nonsingular subspaces. Then, for any G_n, lattices generated by the $\mathscr{M}$'s are studied

by Huo et al. [27–29,32] and lattices generated by the $\mathscr{M}'$'s are studied by Huo and Wan [30,31,33]. Complete solutions of problems (i)–(iii) are obtained except some boundary cases when $G_n = U_n(\mathbb{F}_q)$ $(q = q_0^2)$, $O_{2v+\delta}(\mathbb{F}_q)$ $(n = 2v + \delta)$ and $\mathrm{Ps}_{2v+\delta}(\mathbb{F}_q)$ (q even), which will not be described here.

We again illustrate our study with the symplectic case [27]. Denote by $\mathscr{M}(m,s;2v)$ the orbit of subspaces of type (m,s) in the $2v$-dimensional symplectic geometry over $\mathbb{F}_q$ and by $\mathscr{L}(m,s;2v)$ the lattice generated by $\mathscr{M}(m,s;2v)$. Our main results are as follows:

(i) Assume that $2s \leqslant m \leqslant v+s$ and $2s_1 \leqslant m_1 \leqslant v+s_1$, then $\mathscr{L}(m,s;2v) \supset \mathscr{L}(m_1,s_1;2v)$ if and only if $m - m_1 \geqslant s - s_1 \geqslant 0$.

(ii) Assume that $2s \leqslant m \leqslant v+s$, then $\mathscr{L}(m,s;2v)$ consists of $\mathbb{F}_q^{(2v)}$ and all subspaces of type (m_1,s_1) with $m - m_1 \geqslant s - s_1 \geqslant 0$.

(iii) Assume that $2s \leqslant m \leqslant v+s$ and $m \neq 2v$, then

$$\begin{aligned}
&\chi(\mathscr{L}(m,s;2v),t)\\
&\quad=\sum_{s_1=s+1}^{v}\sum_{m_1=2s_1}^{v+s_1} N(m_1,s_1;2v)(t-1)(t-q)\cdots(t-q^{m_1-1})\\
&\qquad+\sum_{s_1=0}^{s}\sum_{m_1=m-s+s_1+1}^{v+s_1} N(m_1,s_1;2v)(t-1)(t-q)\cdots(t-q^{m_1-1}),
\end{aligned}\tag{36}$$

where $N(m_1,s_1;2v)$ is given by (10).

9. Application to representations of forms by forms

Let us introduce some notations. Let A be an $m \times m$ matrix and B be a $t \times t$ matrix, both over $\mathbb{F}_q$.

(a) When both A and B are alternate, let

$$n^{(a)}_{m\times t}(A,B) = |\{X : m \times t \mid {}^tXAX = B\}|. \tag{37}$$

(b) If q is a perfect square, say $q = q_0^2$, let $a \mapsto \bar{a} = a^{q_0}$ be the involution of $\mathbb{F}_q$, then when both A and B are hermitian, let

$$n^{(h)}_{m\times t}(A,B) = |\{X : m \times t \mid {}^t\bar{X}AX = B\}|. \tag{38}$$

(c) When q is odd, and both A and B are symmetric, let

$$n^{(q)}_{m\times t}(A,B) = |\{X : m \times t \mid {}^tXAX = B\}|. \tag{39}$$

When q is even, let

$$n^{(q)}_{m\times t}(A,B) = |\{X : m \times t \mid {}^tXAX \equiv B\}|, \tag{40}$$

where ${}^tXAX \equiv B$ means that ${}^tXAX - B$ is alternate.

(d) When q is even, and both A and B are symmetric, let

$$n^{(s)}_{m\times t}(A, B) = |\{X : m \times t \mid {}^tXAX = B\}|. \tag{41}$$

(e) Finally, for any $m \times n$ matrix A and $t \times u$ matrix B, both over $\mathbb{F}_q$, let

$$n^{(b)}_{m\times t, n\times u}(A, B) = |\{(X, Y),\ X : m \times t,\ Y : n \times u \mid {}^tXAY = B\}|. \tag{42}$$

Clearly, $n^{(a)}_{m\times t}(A, B)$ is the number of representations of the alternate form with coefficient matrix B by the alternate form with coefficient matrix A, etc.

Assuming that q is odd, in 1954 Carlitz [6,7] obtained closed formula for $n^{(a)}_{m\times t}(A,B)$ and $n^{(q)}_{m\times t}(A, B)$, and in 1955 Carlitz and Hodges [8] obtained closed formula for $n^{(h)}_{m\times t}(A, B)$. Now, using the geometry of classical groups over finite fields, or more precisely, our Anzahl theorems [64], the author [66] obtained closed formula for $n^{(a)}_{m\times t}(A, B)$, $n^{(q)}_{m\times t}(A, B)$, and $n^{(h)}_{m\times t}(A, B)$ without assuming that q is odd. Moreover, closed formulas for $n^{(s)}_{m\times t}(A, B)$ and $n^{(b)}_{m\times t, n\times u}(A, B)$ are also obtained [66].

As an illustration we sketch the deduction of $n^{(a)}_{m\times t}(A, B)$ in steps as follows:

(i) Let

$$A = \begin{pmatrix} K_{2v} & \\ & 0^{(m-2v)} \end{pmatrix}, \tag{43}$$

where K_{2v} is the $2v \times 2v$ nonsingular alternate matrix (6). Then

$$n^{(a)}_{m\times t}(A, B) = q^{t(m-2v)} n^{(a)}_{2v\times t}(K_{2v}, B). \tag{44}$$

(ii)

$$n^{(a)}_{2v\times t}(K_{2v}, B) = \begin{cases} 0 & \text{if } v_1 > v, \\ n^{(a)}_{2v\times 2v_1}(K_{2v}, K_{2v_1}) n^{(a)}_{2(v-v_1)\times(t-2v_1)}(K_{2(v-v_1)}, 0^{(t-2v_1)}) & \text{if } v_1 \leqslant v. \end{cases} \tag{45}$$

(iii) Let $v_1 \leqslant v$, then

$$n^{(a)}_{2v\times 2v_1}(K_{2v}, K_{2v_1}) = q^{v_1(2v-v_1)} \prod_{i=v-v_1+1}^{v} (q^{2i} - 1). \tag{46}$$

(iv)

$$n^{(a)}_{2v\times t}(K_{2v}, 0^{(t)}) = \sum_{k=0}^{\min\{t,v\}} q^{\binom{k}{2}} \frac{\prod_{i=t-k+1}^{t} (q^i - 1) \prod_{i=v-k+1}^{v} (q^{2i} - 1)}{\prod_{i=1}^{k} (q^i - 1)}. \tag{47}$$

Formulas in Steps (i) and (ii) are easy, and formulas in Steps (iii) and (iv) can be deduced from the Anzahl theorems in symplectic geometry. Finally, combining the

formula for $n^{(a)}_{2v\times t}(K_{2v}, 0^{(t)})$ obtained by Carlitz [6] with (iv) we have the following identity:

$$\sum_{k=0}^{\min\{t,v\}} q^{\binom{k}{2}} \frac{\prod_{i=t-k+1}^{t}(q^i-1)\prod_{i=v-k+1}^{v}(q^{2i}-1)}{\prod_{i=1}^{k}(q^i-1)}$$
$$= q^{-\binom{t}{2}+2vt}\,{}_2\Phi_0[q^{-\frac{1}{2}t}, q^{-\frac{1}{2}(t-1)}; q^{v+1}]', \tag{48}$$

where

$${}_2\Phi_0[a, b; z] = \sum_{k=0}^{\infty} \frac{(a)_k(b)_k}{(q)_k} z^k,$$
$$(a)_k = (1-a)(1-qa)\cdots(1-q^{k-1}a) \quad \text{if } k \geqslant 1,$$
$$(a)_0 = 1,$$

and the prime indicates that q is to be replaced by q^{-2} in the function ${}_2\Phi_0$.

10. Concluding remarks

Besides the applications mentioned above it should be added that the geometry of classical groups was also used in the study of correlation properties of binary m-sequences [21,22,34,35].

Classical groups are important algebraic structures and orbits of subspaces are basic geometric objects. Therefore, it is not strange that they have so many applications to the various problems mentioned above and it is natural to hope that further applications will be found in the future.

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